

# Supplementary Material: Context-Dependent Entry Selection in Finite-Rule Tasks (2026-09-10)

Neutral preprint layout — working draft, 10 September 2026.

## S1. Tasks, Actions, and Validation Protocol

The input domain is  $\{-2, -1, 0, 1, 2\}^3$ , containing 125 points. D0 consists of 12 points on the same coordinate plane; the remaining points are divided into 49 ordered evidence points and 64 test points. The 256-member bank is compatible with D0, and the verifier groups semantic hypotheses by their behavior over the full domain. The parent is selected by node count and canonical hash, not by the hidden target.

The program language contains  $x1/x2/x3$ , constants  $-3..3$ , `neg/add/sub/mul`, `gt/eq`, and `ite`. Depth is  $\leq 5$  and node count is  $\leq 31$ ; a classifier must be binary over the full domain. The context library contains 105 motifs: 21 affine commutative, 42 affine directional, 21 affine multiplicative, and 21 pairwise variable motifs. These are context-construction strata, not target classes.

The ten original actions apply five edit frames on each of the two operand paths (left and right) of the parent predicate: `replace`, `add(OLD,CONTEXT)`, `sub(OLD,CONTEXT)`, `sub(CONTEXT,OLD)`, and `mul(OLD,CONTEXT)`. Display option IDs map to the original actions through a frozen mapping. A context must be absent from the parent and appear once in the child. A valid child must agree on D0, be binary over the full domain, differ from the parent, and change evidence behavior. See the source bindings and prompts and strategies.

The four-round procedure and the strong-K4 definition remain in §2.2 of the main text because they affect interpretation of the conclusions and cannot be hidden as appendix-only details. Here we emphasize that the structured oracle returns the first mismatch location/label or `MATCH`; updates filter using the full response rather than a single label. A singleton or an indistinguishable evidence-equivalence state can stop the process early. `K4_full_pool` deduplicates by child behavior within a frozen matching pool with the same parent, frame, motif layer, and complexity; it requires at least three comparison cases, all of which must fail exact identification. It is not the set of all possible contexts. Implementations: `verifier`, `strong-K4`, and `v2` pairing.

## S2. Sample Construction and Version Timeline

From the original pool of 1024 candidates, we construct 24 worlds, six in each of four strata. Each world contributes only one pair, yielding 48 arms. Strict uniqueness is imposed on the successful original action; incorrect actions are not required to have mutually distinct behavior. The two K2 arms are balanced by opportunity count; each nonconstant K4 arm has one correct action, and those actions differ, while constant K4 contributes zero. The v2 configuration fixes selection, display, and the original statistical protocol. The older v1 is retained for history and is not used to reconstruct or rewrite the current results.

A separate full scan of another batch of 1024 candidates produced 19 strict worlds and 41 pairs. Within it, the joint complete-failure class for two nonconstant strategies contained 9 worlds and 22 pairs. We retained all 9 worlds and selected one pair per world by pair hash. The context strata contained 3 affine commutative, 4 affine directional, 2 pairwise variable, and 0 affine multiplicative worlds. Neither the 41 pairs nor the 22 pairs represent the corresponding number of independent worlds; these 9 worlds are not a four-stratum balanced sample. See the search report.

The response and design order was: original non-thinking main experiment → DeepSeek thinking follow-up → nonconstant-strategy diagnostics → GLM follow-up → new-pool strategy challenge → same-nine-world reference controls → same-nine-world rule generation → contemporaneous output-cap control on 8 worlds from the original 24. Freezing each experiment before execution does not mean that every design was preregistered before the original results. Later controls do not independently increase the world count or inherit the original test power.

## S3. All Strategies on the Original 24 Worlds

There are ten fixed-action strategies, ten fixed-display-position strategies, and four public-information strategies, for 24 prespecified strategies. The final two rows are diagnostics proposed after the DeepSeek thinking run and before the GLM run. All use public information. A fixed strategy cannot succeed simultaneously on both arms when their correct actions differ; this is a structural limitation, not evidence that the 20 fixed strategies are 20 independent strong

baselines.

| Strategy ID or diagnostic                                  | own /48 | cross /48 | complete /24 |
|------------------------------------------------------------|---------|-----------|--------------|
| fixed-semantic-00                                          | 14      | 14        | 0            |
| fixed-semantic-01                                          | 0       | 0         | 0            |
| fixed-semantic-02                                          | 0       | 0         | 0            |
| fixed-semantic-03                                          | 2       | 2         | 0            |
| fixed-semantic-04                                          | 12      | 12        | 0            |
| fixed-semantic-05                                          | 12      | 12        | 0            |
| fixed-semantic-06                                          | 0       | 0         | 0            |
| fixed-semantic-07                                          | 0       | 0         | 0            |
| fixed-semantic-08                                          | 1       | 1         | 0            |
| fixed-semantic-09                                          | 7       | 7         | 0            |
| fixed-display-position-00                                  | 4       | 4         | 0            |
| fixed-display-position-01                                  | 4       | 4         | 0            |
| fixed-display-position-02                                  | 7       | 7         | 0            |
| fixed-display-position-03                                  | 9       | 9         | 0            |
| fixed-display-position-04                                  | 3       | 3         | 0            |
| fixed-display-position-05                                  | 7       | 7         | 0            |
| fixed-display-position-06                                  | 2       | 2         | 0            |
| fixed-display-position-07                                  | 6       | 6         | 0            |
| fixed-display-position-08                                  | 5       | 5         | 0            |
| fixed-display-position-09                                  | 1       | 1         | 0            |
| first-public-K1-else-first-displayed                       | 9       | 2         | 2            |
| public-k1-min-node-hash                                    | 22      | 6         | 2            |
| public-k1-min-positive-node-hash                           | 4       | 10        | 0            |
| public-k1-max-parent-novelty-node-hash                     | 29      | 2         | 8            |
| nonconstant filtering + maximum parent-behavior difference | 33      | 1         | 11           |
| nonconstant filtering + minimum nodes                      | 31      | 2         | 8            |

The numbers come from all\_24\_frozen\_policies and separately\_labelled\_nonconstant\_diagnostics in the machine comparison; for the first 24 rows, cross is derived as own – signed\_total. Fixed action IDs select by original action number; fixed display IDs select by position. For the remaining strategies, public-support filtering, positive-node-change handling, and tie rules follow the implementation; diagnostic definitions are given in the case record. In the 9-world challenge, all 26 strategies had complete success of 0/9, but selection was conditioned on two focal strategies and must not be treated as 26 independent validations.

#### S4. Per-World Results for the New Observation and Scoring Boundaries

There was one independent response per condition, using the same targets and 64 private test points. The table follows the frozen ordinal rather than sorting by performance. The total correct counts for no additional observation, new observation, and repeated old observation were 305/399/379, respectively, with denominator 576 in each case. Complete test recovery and full 125-point recovery were both 0/9 for all groups.

| World ordinal | no additional correct /64 | new observation correct /64 | repeated observation correct /64 |
|---------------|---------------------------|-----------------------------|----------------------------------|
| 0             | 12                        | 32                          | 49                               |
| 1             | 13                        | 45                          | 46                               |
| 2             | 39                        | 39                          | 39                               |
| 3             | 42                        | 51                          | 51                               |
| 4             | 35                        | 57                          | 30                               |
| 5             | 47                        | 47                          | 47                               |
| 6             | 40                        | 51                          | 40                               |
| 7             | 34                        | 34                          | 34                               |
| 8             | 43                        | 43                          | 43                               |

After the new observation, the remaining candidate counts were 189, 187, 161, 184, 86, 171, 89, 85, and 61, in order. Only 2/9 contradicted the parent; all were retained. New minus repeated gave 2 wins, 2 losses, and 5 ties, so it is not valid to show only worlds 4 and 6. The repeated points had already been seen, so this control does not match the new observation in novelty, input position, or strict information amount. See the new-observation results and design for full scoring and code baselines. Programs that were valid but did not match the observation still contribute their original test accuracy, with consistency reported separately. All 27 trials in this round were valid and consistent; none was removed by consistency.

## S5. Statistics, Parameters, and Technical Record

The original primary test used a one-sided exact sign test on the net utilization direction among non-tie worlds, with  $\alpha=0.05$ . Original non-thinking produced 0 favorable, 1 adverse, and 23 tie outcomes, giving  $p=1$ . When invalid content was received, the entire world was recorded as a tie at the primary endpoint and as a complete failure. Transport failures or missing responses made the entire round unevaluable rather than contributing a partial denominator. All 48 original responses were valid. Each subsequent experiment followed its own technical-failure rule, did not retry based on correctness, and was not retroactively treated as a reproduction of the original primary test.

Original DeepSeek non-thinking used temperature 0.2 with a cap of 256. DeepSeek thinking high used a cap of 131072 without explicitly sending temperature/top\_p. GLM thinking high used a cap of 131072, temperature 1.0, and top\_p 0.95. The subsequent challenge, reference controls, and rule generation reused the DeepSeek thinking configuration. The contemporaneous output-cap controls used non-thinking, temperature 0.2, and caps of 256 or 8192, with all other request contents fixed. Each task had an independent context; a provider’s high setting or a shared cap does not imply equal computation.

| Model experiment                | final responses | physical task attempts | known output tokens |
|---------------------------------|-----------------|------------------------|---------------------|
| Original DeepSeek non-thinking  | 48              | 48                     | 469                 |
| Original DeepSeek thinking      | 48              | 49                     | 736187              |
| Original GLM thinking           | 48              | 49                     | 1039407             |
| 9-world challenge               | 18              | 18                     | 379301              |
| Three-condition reference       | 54              | 54                     | 911963              |
| Three-condition rule generation | 27              | 27                     | 308880              |
| Two non-thinking output caps    | 32              | 33                     | 316                 |

The table excludes technical prechecks. Usage for failed attempts is partly unknown, so these figures are not used to calculate a complete bill or an exact total cost. Provider-counted reasoning included in output is not added again; a missing reasoning field does not mean that internal computation was zero. All final responses were valid and were not truncated. For the output-cap control, known successful input was 21364 and output was 316, for a total of 21680; timeout consumption is unknown. Per-condition usage for the reference controls and rule generation is reported in the respective result reports.

## S6. Reproduction Material Entry Points and Release Boundary

| Evidence                        | Protocol / method | Completed results and verification entry point |
|---------------------------------|-------------------|------------------------------------------------|
| Original 24 worlds              | v2 configuration  | original primary results, DS thinking, GLM     |
| Original 4/5-round code checks  | budget script     | results JSON                                   |
| New-pool challenge              | call plan         | model results                                  |
| Auxiliary reference             | design, protocol  | results                                        |
| New-observation rule generation | design, protocol  | results                                        |
| Non-thinking output caps        | protocol          | results                                        |

Each result report points to the local artifacts directory containing the plan, private mapping, generation, analysis, results, and raw response/attempt records, and lists the corresponding hashes and completed read-only checks. The exact original prompts, per-world mappings, and remaining pair statistics follow these frozen files and are not manually reconstructed in this supplement.

Historical reports' "not executed later" statements reflect their earlier status only; the current top-level HANDOFF governs.

Reproduction does not require sending new model requests. Do not rerun run/freeze/build in a way that overwrites old artifacts. Model raw responses, private targets, and key files must not be combined into one publishable directory. This supplement contains no keys and does not imply authorization to publish private artifacts. Before submission, the distributable file list still needs to be determined, a credential-free reproduction package prepared, the exact prompts added as an appendix, and an independent-environment validation completed. These remain distinct from the current local audit passing.